FAROUK MSELMI

Professional Address
Building B4, Office 2.39
Euro-Mediterranean University
Roundabout Bensouda
Meknes Road, Fez, Morocco

Personal Address
Apartment 25, Prestige 2 Residence
17 Abou Baker Zniber street
V.N. Champ de course, Fez, Morocco

PERSONAL INFORMATION

Gender: Male.

Date and place of birth: January 20, 1986 in Ain Drahem city, Tunisia.

Citizenship: Tunisian.

Family situation: Married, one son, one daughter.

Phone number: +212611535245.

AFFILIATIONS, EMAILS AND LINKS

Affiliation: Euro-Mediterranean University, Morocco.
E-mails: mselmi.farouk@gmail.com; f.mselmi@ueuromed.org
Researchgate: https://www.researchgate.net/profile/Farouk_Mselmi
ORCID: https://orcid.org/0000-0003-1881-7046

RESEARCH AREAS

Generalized Linear Models; Mathematical Statistics; Kernel Estimation Method; Nonparametric Statistics; Mixture Models; Natural Exponential Families; Variance Functions; Stable Processes.

PROFESSIONAL EXPERIENCE

Associate professor at Euro-Mediterranean University, Fez, Morocco.	Sep. 2026 - Present
Assistant professor at Euro-Mediterranean University, Fez, Morocco.	Jan. 2021 - Sep. 2026
Assistant professor at King Saud University, Riyadh, Saudi Arabia.	Sep. 2018 - Jan. 2021
Assistant professor at Sfax University, Sfax, Tunisia.	Feb. 2016 - Sep. 2017
Teaching assistant at Sfax University, Sfax, Tunisia.	Oct. 2015 - Dec. 2015
Teaching assistant at Gabes University, Gabes, Tunisia.	Oct. 2011 - Sep. 2015

DIPLOMAS AND CERTIFICATES

French qualifications for 'Maître de Conférences' and 'Professor' in section 26 of the CNU, 2024
qualification numbers 'PR: 24126372697' and 'MCF: 22226372697'.

Habilitation thesis (High Degree of Research) from Burgundy-Franche-Comté University (new name: Marie and Louis Pasteur University), France, (In French language: Habilitation à Diriger des Recherches (HDR)).
Title: Markov Kernels Models for Natural Exponential Families. Apr. 3, 2023
Jury members: Pr. Célestin Kokonendji (President, Marie and Louis Pasteur University, France),
Pr. Jean-François Dupuy (Reviewer, Rennes University, France),
Pr. Abderrazed Karoui (Reviewer, Carthage University, Tunisia),
Pr. Yousri Slaoui (Reviewer, Poitiers University, France),
Pr. Mahdi Louati (Examiner, Sfax University, Tunisia),
Pr. Angelo Koudou (Examiner, Lorraine University, France).
https://theses.hal.science/tel-04069084/

Ph.D in mathematics (Probability and Statistics) at Sfax University, Tunisia.
Title: Multivariate normal α-stable distributions. Sep. 2012 to Dec. 18, 2015
Advisor: Afif Masmoudi, Co-Advisor: Mahdi Louati.
Jury members: Pr. Mokhless Hammami (President, Sfax University, Tunisia),
Pr. Sonia Fourati (Reviewer, Rouen Normandy University, France),
Pr. Mohamed Ali Hammami (Reviewer, Sfax University, Tunisia),
Pr. Bassem Ben Hamed (Examiner, Sfax University, Tunisia).
https://www.researchgate.net/publication/324775442_Multivariate_Normal_a-Stable_Distributions

Master degree in mathematics and applications at Tunis El Manar University, Tunisia.
Title : Large deviations for local time and intersection local time of fractional Brownian motion and Riemann-Liouville process. Sep. 2009 to Jul. 14, 2011
Advisor: Ezeddine Haouala.

Bachelor degree in mathematics at Carthage University, Tunisia. Sep. 2004 to Jun. 2008

High school diploma in mathematics at Ain Drahem high school, Tunisia. Jun. 2004

PUBLICATIONS

Peer-Reviewed Journal Articles

1. F. Mselmi, *Regression models for composite responses with heterogeneous distributional components.* Sankhya A (Springer). DOI: https://doi.org/10.1007/s13171-026-00453-w. 2026.
https://link.springer.com/article/10.1007/s13171-026-00453-w

2. F. Fourati, F. Mselmi, N. Limnios, *Asymptotic Thorin distributions of multivariate generalized splines via Markov–Krein Transform.* Journal of Statistical Theory and Practice (Springer). Vol 30, 62, 1-23. 2026.
https://link.springer.com/article/10.1007/s42519-026-00571-6

3. F. Mselmi, *On the duality for mixture models.* Statistics and Probability Letters (Elsevier). Vol 230, 110604, 1-5. 2026.
https://www.sciencedirect.com/science/article/pii/S0167715225002494

4. F. Mselmi, C.C. Kokonendji, *Multivariate generalized linear models for Markov kernels with (un)known link and variance functions.* Test (Springer), Vol 34, No 4, 896-926. 2025.
https://link.springer.com/article/10.1007/s11749-025-00982-w

5. F. Mselmi, M. Fri, N. Rouky, *Urban Mobility Modeling: Application to Seoul Bike Sharing Data.* Engineering Proceedings (MDPI). Vol 112, No 51, 1-9. 2025.
https://www.mdpi.com/2673-4591/112/1/51

6. S. Charfi, F. Mselmi, *Modeling exchange rate volatility: application of GARCH models with a Normal Tempered Stable distribution.* Quantitative Finance and Economics (AIMS Press). Vol 6, No 2, 206-222. 2022.
https://www.aimspress.com/article/doi/10.3934/QFE.2022009

7. F. Mselmi, *Generalized linear model for subordinated Lévy processes.* Scandinavian Journal of Statistics (Wiley). Vol 49, No 2, 772-801. 2022.
https://onlinelibrary.wiley.com/doi/10.1111/sjos.12538

8. F. Mselmi, *Approximation of the quasi-deviance function for the time-changed Lévy processes by the first-exit time of the inverse Gaussian subordinator.* Stat (Wiley), Vol 10, No 1, 1-9. 2021.
https://onlinelibrary.wiley.com/doi/10.1002/sta4.362

9. M. Louati, A. Masmoudi, F. Mselmi, *The normal tempered stable regression model.* Communications in Statistics: Theory & Methods (Taylor & Francis), Vol 49, No 2, 500-512. 2020.
https://www.tandfonline.com/doi/full/10.1080/03610926.2018.1554121

10. F. Mselmi, *Lévy processes time-changed by the first-exit time of the inverse Gaussian subordinator.* Filomat (Faculty of Sciences and Mathematics, University of Nis), Vol 32, No 7, 2545-2552. 2018.
http://www.pmf.ni.ac.rs/filomat-content/2018/32-7/32-7-16-6205.pdf

11. F. Mselmi, C.C. Kokonendji, M. Louati, A. Masmoudi, *Generalized variance functions for infinitely divisible mixture distributions.* Mediterranean Journal of Mathematics (Springer), Vol 14, No 4:165, 1-19. 2018.
https://link.springer.com/article/10.1007/s00009-018-1214-9

12. F. Mselmi, *Characterization of the inverse stable subordinator.* Statistics and Probability Letters (Elsevier), Vol 140, 37-43. 2018.
https://www.sciencedirect.com/science/article/pii/S0167715218301470

13. F. Mselmi, *The stable processes on symmetric matrices.* Arab Journal of Mathematical Sciences (Elsevier), Vol 24, No 1, 63-69. 2018.
https://www.sciencedirect.com/science/article/pii/S1319516617300841

14. M. Louati, A. Masmoudi, F. Mselmi, *Characterizations of multivariate stable processes.* Lithuanian Mathematical Journal (Springer), Vol 57, No 1, 59-68. 2017.
https://link.springer.com/article/10.1007/s10986-017-9343-6

15. M. Louati, A. Masmoudi, F. Mselmi, *Multivariate normal α-stable exponential families.* Mediterranean Journal of Mathematics (Springer), Vol 13, No 3, 1307-1323. 2015.
https://link.springer.com/article/10.1007/s00009-015-0562-y

Books

1. Accepted for publication: C.C. Kokonendji, F. Mselmi. Compound Tweedie Dispersion Models: Theory, Methods, and Applications. Chapman & Hall / CRC Press (Taylor & Francis Group), London, pp. 1-302. 2026.

Peer-Reviewed Chapters/Conference Papers

1. S. Charfi, F. Mselmi, *Stock Market Volatility Forecasting: A Supervised Learning Approach with Normal Tempered λ-Stable Distribution.* 2024 3rd International Conference on Embedded Systems and Artificial Intelligence (ESAI), 1-5 (IEEE Xplore). 2024.
https://ieeexplore.ieee.org/document/10913516

2. N. Rouky, M. Fri, F. Mselmi, *Comparative performance analysis for Poisson and Poisson-Inverse Gaussian generalized linear models on supply chain management.* Lecture Notes in Networks and Systems, 1104 LNNS, 91-98 (Springer). 2024.
https://link.springer.com/book/10.1007/978-3-031-68628-3

3. M. Louati, A. Masmoudi, F. Mselmi, *Gamma stopping and drifted stable processes.* Springer Proceedings in Mathematics and Statistics, 131, 223-232 (Springer). 2015.
https://link.springer.com/chapter/10.1007/978-3-319-18041-0_13

SUBMISSIONS

1. Journal article: F. Mselmi, S. Ferreira, *Generalized estimating equations for longitudinal data with inverse stable margins.* 2026.

2. Journal article: F. Mselmi, *Distribution-Aware Hybrid Neural Learning for Flexible Count Prediction: A Poisson–Tempered Stable Framework.* 2026.

3. Journal article: F. Mselmi, C.C. Kokonendji, *Computational Inference for Poisson–Tweedie Regression with Data–Driven Power Selection.* 2026.

4. Journal article: F. Mselmi, *Lévy structure of dual measures for real mixture models.* 2026.

MEMBERSHIPS IN SCIENTIFIC COMMITTEES

- December 4 to 5, 2026: Member of Scientific Committee for the 2nd International Conference on Artificial Intelligence and Cognitive Science for Emerging Technologies (AICSET 2026), Settat, Morocco.
https://aicset.org/committees.

- July 14 to 16, 2025: Member of Scientific Committee for the International Conference on Artificial Intelligence and Cognitive Science for Emerging Technologies (AICSET 2025), Marrakesh, Morocco.
https://aicset.sciencesconf.org/resource/page/id/5.

- December 02 to 06, 2024: Member of Scientific Committee for the International Conference on the Promotion of Applied Statistics (CIPSA'2024) for decision-making and development in Africa, Yaoundé, Cameroon.
- May 2 to 4, 2024: Chair of the thematic session titled "Modeling, simulation and optimization" during the 7th International Conference on Logistics Operations Management, Marrakesh, Morocco.
- May 2 to 4, 2024: Member of the editorial committee for the special session: "Artificial Intelligence Techniques and Statistical modeling for Mobility and urban logistics planning" The 7th International Conference on Logistics Operations Management, Marrakesh, Morocco.
https://gol24.sciencesconf.org/resource/page/id/31.

- October 14 to 16, 2016: Member of the organizing committee of the 3rd International Conference on Bayesian Networks and Applications, Sfax University, Tunisia.

OTHER PROFESSIONAL ACTIVITIES

Referee of papers for: Journal of Statistical Theory and Practice (Springer)(1), Metrika (Springer)(1), Communication in Statistics: Simulation and Computation (Taylor & Francis)(1), Communication in Statistics: Theory and Methods (Taylor & Francis)(1), Annals of the Institute of Statistical Mathematics (Springer)(1), Statistical Papers (Springer)(1), The R Journal(1).

PARTICIPATION IN FUNDED RESEARCH PROJECTS

- 2022-2025: Urban logistics: systemic approach, modelling and practical solutions - The case of Mediterranean conurbations.

  Budget: 511.032 kDh from Euromed University.

- 2021-2024: Urban mobility e-marketplace for optimized travel (Mobility Online Market for Optimized Trips).

  Budget: 1636 kDh from Ministry of Higher Education and CNRST with ADD.

CONFERENCES AND WORKSHOPS

- October 15 to 16, 2026: Invited speaker in the 6th International Conference on Statistics, Mathematics and Computation–WSMC16. Organized by Universidade Aberta and Universidade da Beira Interior, Portugal. Oral presentation entitled: "Regression models for composite responses with heterogeneous distributional components".

- April 23 to 25, 2026: Participation in The 8th Edition of the International Conference on Research in Applied Mathematics and Computer Science (ICRAMCS 2026), Marrakech, Morocco.

- July 14 to 16, 2025: Participation in The International Conference on Artificial Intelligence and Cognitive Science for Emerging Technologies (AICSET 2025), Marrakech, Morocco. Oral presentation entitled: "Urban Mobility Modeling: Application to Seoul Bike Sharing Data".

- December 19 to 20, 2024: Participation in The 3rd International Conference on Embedded Systems and Artificial Intelligence (ESAI'24), Fez, Morocco. Oral presentation entitled: "Stock Market Volatility Forecasting: A Supervised Learning Approach with Normal Tempered Stable Distribution".

- December 2 to 6, 2024: Participation in the International Conference on the Promotion of Applied Statistics for decision-making and development in Africa CIPSA'2024, Yaounde, Cameroon. Oral presentation entitled: "Modeling the ratio of Tanimoto coefficients by the normal tempered stable model".

- May 6 to 4, 2024: Participation in the 7th edition of the International Conference on Logistics Operations Management, GOL'24, Marrakech, Morocco. Oral presentation entitled: "Comparative performance analysis for Poisson and Poisson-Inverse Gaussian generalized linear models on supply chain management".

- July 6 to 8, 2022: Participation in the 3rd LmB Conference on Multivariate Statistical Models : Count and Semi-Continuous, Besançon, France. Oral presentation entitled: "Generalized linear model for subordinated Lévy processes".

- February 26 to March 1, 2017: Participation in the 22nd international conference of Tunisian Mathematical Society (TMS), Mahdia, Tunisia. Oral presentation entitled: "Characterizations of the normal variance-mean mixture by a stable mixing".

- February 9 to 11, 2017: Participation in workshop of Modeling, Statistical Signal and Remote Sensing, Sfax, Tunisia.

- October 14 to 16, 2016: Participation in the 3rd International Conference on Bayesian Networks and Applications (ICBNA), Sousse, Tunisia.

- December 21 to 24, 2015: Participation in the International Conference on Advances of Applied Mathematics (ICAAM), Hammamet, Tunisia. Oral presentation entitled: "Multivariate normal α-stable distributions".

- December 22 to 25, 2014: Participation in the International Conference on Advances of Applied Mathematics (ICAAM), Hammamet, Tunisia. Oral presentation entitled: "Multivariate normal α-stable exponential families".

- December 16 to 19, 2013: Participation in the International Conference on Advances of Applied Mathematics (ICAAM), Hammamet, Tunisia. Oral presentation entitled: "Gamma stopping time and drifted stable processes".

- October 28 to November 9, 2013: Participation in "CIMPA-School", Nabeul-Hammamet, Tunisia.

SCIENTIFIC VISIT

Following an invitation from Ms. Sonia Fourati, a scientific visit was made to INSA Rouen, France, for the period from June 15 to July 01, 2015. An oral presentation entitled: "Multivariate normal α-stable distributions" was made on June 25, 2015.

MEMBERSHIPS IN ADMINISTRATIVE COMMITTEES

- 2024: Participation in the accreditation of the industrial engineering program, Euro-Mediterranean University, Fez, Morocco.

- 2022 to 2024: Coordinator of several teaching modules for preparatory and engineering cycles.

- 2024: Participation in the accreditation of the Master's in Supply Chain Management program, Euro-Mediterranean University, Fez, Morocco.

- 2022-2023: Participation in the doctoral selection process, Euro-Mediterranean University, Fez, Morocco.

- 2022-2023: Participation in the selection of students in the Master's in Transport and Sustainable Mobility, Euro-Mediterranean University, Fez, Morocco.

- Since 2022: Member of the Euromed Polytechnic School Council, Euro-Mediterranean University, Fez, Morocco.

- 2021: Member of the coordination committee for the preparatory cycle for engineering studies, Euro-Mediterranean University, Fez, Morocco.

COMPUTING SKILLS

Matlab; Octave; Python; R; C; C++; LaTeX.

LANGUAGE

Arabic: Native; English: Fluent; French: Fluent.